

Electronic Companion

Exact Path Representations for Finite-Catalog Renewal Design

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This companion distinguishes current diagnostics from retained antecedent proofs. The full historical sources and readers remain unchanged. Counts in the retained resource-checker discussion describe its antecedent validation, not the current 130-model test.

EC.1 Current Execution Scope and Detailed Diagnostics

The current R54 study is distinct from the inherited R52 and R53 experiments. Its original protocol preceded the regression and 142-method comparison. A later addendum records the SUBSET SUM reduction and the heterogeneous component-packing extension, both derived during this revision. The addendum preceded the reduction checks and four structural challenge runs. Packing checks use the already declared regression models and are identified as post-derivation development checks, not a new unseen sample. The three exact sharpness neighbors are also separate development examples. No previous result is overwritten by these additions.

The resource cap is enforced as a process address-space limit of 2 GiB. Resident memory is an outcome, measured by the operating system, and is not claimed to be identically allocated across methods. Each method is run in its own process. Scientific imports are timed before the algorithm alarm and reported separately. Certificates are serialized after the algorithm stage, then independently verified. Time-limited compiled numerical routines can defer a Python signal; the parent process imposes an additional algorithm-phase wall-clock cutoff. A numerical solve that returns after an alarm is not silently counted as within the requested budget. The raw records contain actual times, the last completed phase, and process-limit statuses.

For an interrupted enumeration, completed books may improve the feasible incumbent, but unvisited books require the universal upper bound. For an interrupted uniform or class-box routine that has not returned a complete witness, a fallback original feasible policy is retained. Those wide fallback intervals are not advertised as the internal performance of a partially executed optimizer. In contrast, the new method keeps a complete cover throughout and returns its latest valid price-tree bound after a controlled interruption. Every source-model digest, requested tolerance, method limit and final evidence scope is included in the records.

Table EC.1 separates actual numerical solver time, solver time allowance, analytical envelope error, and the signed numerical bracket width. The machine-readable records also retain every probability, service assignment, installed support, primal and dual value, status, relative solver gap and maximum constraint residual. Small negative floating-point widths are reported as roundoff, not rounded into an exact-proof claim. The new checker uses rational arithmetic and explicit exceptions; the old grid and box checkers retain their documented requirement not to disable assertions.

Table EC.2 gives every target-command threshold slack in the screening benchmark. For each of the eight commands in each case, the raw record additionally gives exact inclusion/exclusion values, the two safe-test decisions, possible-anchor status and whether the command is wholly ineligible. An omitted command from one displayed winning book need not be irrelevant: the inclusion reference determines whether some equally

good book uses it. Strict irrelevance means its forced optimum is strictly below the global reference; the generated diagnostics count each test's missed strictly irrelevant commands. The benchmark does not claim to recover a mixed-integer solver's internal presolve-removal rate. It compares a cheap uniform deletion rule with stronger original-instance price fixing and with exact relevance.

The sharpness paragraph in the retained anchor-envelope proof requires a feasible singleton and pair, $m \geq 2$, and $B = \bar{b}$. Under those conditions, the target vector is fixed at all caps and the pair-minus-singleton gross difference equals the envelope. Without them, the envelope remains safe but need not be attained at the accepted promise. Table EC.3 contains three exact neighbors of such a configuration. All have the same nonuniform caps and reward/service primitives; changing only the opening charge changes necessity, a tie, and strict exclusion. This resolves the distinction between safety of the equality test and exclusion from every optimum.

For context, the retained R53 controlled screen used width 1/10 and transferred the interval $[15/64, 91/320]$. Its width is 1/20, or $16/75 \approx 21.33\%$ of the reference 15/64. Its exact optimum was supplied by enumeration, not by that positive-width interval. Those historical cases and their wholly ineligible endpoint removal remain intact; they are not substituted for the present near-threshold evaluation.

EC.2 Independent Production Configuration Example

The retained production theorem applies to a compatibility graph whose selected modules share divisible load in parallel. For the current numerical example, the capacity potentials are (0, 2, 5, 9) and all increasing vertex pairs are compatible. For edges (0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3), the respective triples of linear return, quadratic curvature, and opening payment are

$$(5, 1, 1/2), (4, 1/2, 1), (3, 1/2, 0), (7, 2, 3/2), (9/2, 1, 1/4), (6, 3/2, 3/4).$$

Each edge's capacity is the difference of its endpoint potentials. The accepted orders are 3, 6, 8, crossed with module budgets 1, 2, 3. Every feasible configuration is enumerated; clipped stationary allocations and a common rational supporting price prove its exact operating optimum. Table EC.3 displays all nine optimum configurations. Full allocations, all alternative paths and exact values are retained. These module-specific returns and capacity coordinates were specified independently, with no mapping from renewal histories or terminal lotteries. The example demonstrates the capacity-conservative abstraction numerically; it does not assert an observed production installation or estimated economic parameters.

EC.3 Verification of Original-Instance Binding

Price certificates use a canonical rational model representation and SHA-256 of its sorted JSON encoding. A verifier can supply the expected digest from the requested model record. It then checks mathematical feasibility and all price-tree bounds, not merely the digest. Different encodings must first be converted to the documented canonical representation; a certificate's self-declared hash does not authenticate an external model. The budget is retained as supplied, including budgets exceeding the catalog size, while the recurrence

clips its effective command count. A separate regression checks that normalization does not silently change this instance binding.

The singleton/pair certificate contains every feasible singleton and pair permitted by the budget and a fixed-book supporting price for each. Its independent checker evaluates the original scalar responses at eligible knots, individual caps and clipped stationary tail points. Their supporting bounds, together with one original feasible policy attaining their maximum, establish the exact value under the common-cap theorem. No grid, search-tree completeness flag or numerical status is used as a substitute for this coverage.

EC.4 Joint Optimization with One Resource Grid

EC.4.1 A capacity-repair principle

We first state the approximation argument for a general resource-path problem. It identifies exactly which property is supplied by Theorem 3.4.

Consider a finite acyclic directed graph with designated source and sink. An admissible path has at most h arcs. Arc e has capacity $C_e \geq 0$, opening charge κ_e , and an L -Lipschitz payoff ψ_e on $[0, C_e]$. Choose a path and resources satisfying $\sum_e x_e = R$, to maximize $\sum_e [\psi_e(x_e) - \kappa_e]$. Assume every admissible path has total capacity at least $R \geq 0$. This assumption is verifiable by a minimum-capacity path calculation with the same length restriction; it is automatic when $C_{uv} = p_v - p_u$ for a nondecreasing vertex potential and $R \leq p_{\text{sink}} - p_{\text{source}}$.

For $\eta > 0$, restrict resources to $x_e = q_e \eta$, where q_e is a nonnegative integer and $q_e \eta \leq C_e$. Accept a grid path only if

$$0 \leq R - \eta \sum_e q_e \leq h\eta. \quad (1)$$

Let S_η be the largest grid-path score in this band.

Theorem EC.4.1 (Capacity-repair certificate). *A maximizing grid path can be repaired to exact resource R without changing its arcs. If its repaired feasible value is $\underline{V}$, the optimum V satisfies*

$$S_\eta - Lh\eta \leq \underline{V} \leq V \leq S_\eta + Lh\eta. \quad (2)$$

Consequently the returned path is within $2Lh\eta$ of optimal. The result permits signed payoffs and arbitrary fixed charges; it does not use a positive lower bound on the optimal value.

Proof. Round each resource on an optimal path down to a multiple of η . The lost resource d is less than $h\eta$, so the rounded vector satisfies (1). Lipschitz continuity bounds its objective loss by Ld , because charges do not change. Hence $V \leq S_\eta + Lh\eta$.

For the maximizing grid path, its unused total capacity is at least $R - \eta \sum_e q_e$. Distribute exactly that residual among its nonnegative remaining capacities. No coordinate decreases and the sum of absolute resource changes equals the residual, which is at most $h\eta$. The payoff loss is at most $Lh\eta$. The repaired vector is feasible for the original resource requirement and uses the same path. This proves all inequalities. $\square$

Concavity is not needed for this error argument. It will give efficient convolution and reconstruction in the renewal problem. An arbitrary resource-constrained path problem with some accepted grid paths of insufficient total capacity would not satisfy this theorem; its rounded solutions could be infeasible to repair. The identity (11) rules out precisely that failure here.

EC.4.2 Renewal recurrence and exact-obligation recovery

For each feasible catalog anchor $a_h \leq \min(B, \min_j b_j)$, set $R_h = B - a_h$ and $Q_h = \lfloor R_h/\eta \rfloor$. Let $D_{\ell,v}^h(z)$ be the best score of a path that starts at a_h , ends at a_v , has ℓ selected commands, and consumes resource $z\eta$ on its nonterminal edges. The initial condition is

$$D_{1,h}^h(0) = f(a_h) - \rho_h.$$

Unreachable states have value $-\infty$. For $\ell \geq 2$,

$$D_{\ell,v}^h(z) = \max_{\substack{u < v, 0 \leq t \leq z \\ (z-t)\eta \leq T_{uv} + Q_{uv}}} \{D_{\ell-1,u}^h(t) + \Phi_{uv}((z-t)\eta) - \rho_v\}. \quad (3)$$

Complete each state with its terminal edge. Retain totals z in the band

$$\max \left\{ 0, \left\lceil \frac{R_h - m\eta}{\eta} \right\rceil \right\} \leq z \leq Q_h. \quad (4)$$

The maximum over all anchors, all $1 \leq \ell \leq m$, and all terminal completions is S_η . The terminal edge counts among the at most m rounded edges. It does not consume another command slot or incur another command charge.

Theorem EC.4.2 (Polynomial additive finite-catalog design). *Under (4), let the inputs and charges be rational, and choose $\eta = \varepsilon/(2Lm)$ for $\varepsilon > 0$. Recover an optimal fixed-book policy at the original promise B from a maximizing grid path. Its value $\underline{V}$ and the bound $\bar{V} = S_\eta + Lm\eta$ satisfy*

$$\underline{V} \leq V^*(B) \leq \bar{V}, \quad 0 \leq \bar{V} - \underline{V} \leq \varepsilon. \quad (5)$$

All original individual caps, realization ceilings, opening charges, and the command budget are respected. With $Q = \lceil 2Lm/\varepsilon \rceil + 1$, a direct exact implementation uses

$$O\left(kN^2Q \log(k+1) + mN^3Q \log(Q+1)\right) \quad (6)$$

arithmetic operations. Its bit complexity is polynomial in k, N, m, Q and the binary input length. Neither count has an exponent depending on the number of eligibility classes.

Proof. For every feasible anchor, Theorem 3.4 gives capacity $\bar{b} - a_h \geq R_h$ on every path. Its arc functions have Lipschitz constant L . Apply Theorem EC.4.1 with at most m edges, including the terminal edge. A repaired resource solution can be implemented with no smaller payoff by the rearrangement in Theorem 3.4.

Exact fixed-book optimization at B can only improve this feasible reconstruction. Every optimal original book has an enumerated feasible anchor, so maximizing across anchors gives (5).

There are $O(N^2)$ possible edges, including terminal edges. Each has at most Q grid values. Solving its capped quadratic allocation by sorting at most k marginal breakpoints costs $O(k \log(k + 1))$ operations per value; this conservative preprocessing bound matches the implemented reference construction. The monotone convolution below evaluates one edge transition in $O(Q \log(Q + 1))$ operations. There are at most N anchors and mN^2 transitions per anchor. Terminal completions are included in the same bound.

Every grid value is rational with numerator and denominator length polynomial in the input length and $\log Q$. The least common multiple of the finitely many grid denominators, anchor rewards, and charges has bit length no greater than the sum of those lengths, hence polynomial in k, N, Q and input length. Scaling by this *exact* denominator converts all Bellman additions and comparisons to integer arithmetic without payoff quantization. Summing at most m terms preserves polynomial bit length. Recovery and the final bound use rational arithmetic as well. $\square$

For normalized accuracy $\epsilon = \varepsilon/L$, $Q = O(m/\epsilon)$. Thus the theorem is polynomial in inverse normalized accuracy even when L is encoded in binary. For a prescribed absolute error, the dependence is on the numerical ratio L/ε , not on its logarithm. This is an additive scheme, not a relative-error fully polynomial approximation scheme. Large opening charges or an optimum near zero do not invalidate the certificate.

EC.4.3 Concave kernels, not concave Bellman values

The sequence $K_e(q) = \Phi_e(q\eta)$ is concave. For an arbitrary previous-state vector $A(t)$, the transition matrix

$$M(z, t) = A(t) + K_e(z - t)$$

is anti-Monge on its feasible band: the column terms cancel, and decreasing successive differences of K_e give the quadrangle inequality. Its leftmost row maximizers are nondecreasing. A divide-and-conquer row search therefore evaluates the max-plus convolution in $O(Q \log(Q + 1))$ comparisons. The previous vector need not be concave. All reachable resource indices form a prefix, because each edge admits every integer from zero to its capacity index; taking the maximum over paths preserves this contiguous support.

This distinction is necessary. With one history, cap one, zero service cost, $f(x) = x$, catalog $\{0, 1/2\}$, budget two, and charges $\rho(0) = 0$, $\rho(1/2) = 1/10$, the optimized value is

$$V^*(B) = \max\{0, \min(B, 1/2) - 1/10\}, \quad 0 \leq B \leq 1.$$

Its slope increases at $B = 1/10$, so it is not concave. A single supporting price need not certify the global menu choice. The resource grid avoids that incorrect concavity inference.

EC.4.4 Independent global certificates

A resource certificate contains the original model, grid step, common denominator, all feasible-anchor Bellman tables, the best band score, and an original-space lottery policy. The checker imports neither the

new optimizer nor an antecedent optimizer. It reconstructs service values by enumerating marginal-price intervals. For an interval with saturated cost c_0 , saturated mass y_0 including free service, and active reciprocal curvature $R_0 = \sum_{j \in A} \pi_j / \gamma_j$, it uses

$$\alpha = (y - y_0) / R_0, \quad H(y) = c_0 + \frac{1}{2} R_0 \alpha^2,$$

checks that α lies in the stated interval, and evaluates every arc grid value exactly. Free service and zero capacities are handled separately.

It then checks every Bellman row using totally monotone matrix search (Aggarwal et al., 1987), rather than the optimizer's divide-and-conquer recurrence evaluator. For a finite kernel and previous vector, a sufficiently steep concave linear continuation outside the kernel support makes all infeasible entries strictly inferior to a feasible one. This permits ordinary rectangular matrix search without treating equal negative-infinity entries as valid witnesses. The checker verifies the complete anchor and state sets, the terminal band maximum, and $\bar{V} = S_\eta + Lm\eta$. Finally, it checks nonnegative lottery probabilities, unit mass, all realized ceilings, individual expected caps, exact root equality, the selected-book charge, and the stated interval width. A completion flag is never used as evidence.

Storage is $O(mN^2Q + N^2Q)$ rational or scaled-integer entries when all anchor tables are retained; an optimizer that does not retain certificates can process anchors sequentially. Verification cannot be sublinear in the supplied certificate size. The experiments report both storage and checking costs rather than counting only the command alphabet.

EC.5 Exact Common Eligibility and Catalog Stability

EC.5.1 The exact one-class result remains available

For the full catalog, define $K_j = |\mathcal{A} \cap [0, \tau_j]|$. Histories with equal K_j form a catalog-eligibility class G_g ; let E be their number, $W_g = \sum_{j \in G_g} \pi_j$, and $w_{j|g} = \pi_j / W_g$. These classes are sufficient for a class-mean algorithm, not an intrinsic dimension of every optimal selected book.

Fix one such class and a common eligible selected prefix with last level d . At mean s , define capped equalization $\theta_j(s) = \min(b_j, z)$, where $\sum_{j \in G_g} w_{j|g} \min(b_j, z) = s$. Let

$$D_g(d) = \sum_{j \in G_g} w_{j|g} \min(b_j, d), \quad C_g(y; d) = \min_{\substack{0 \leq z_j \leq (b_j - d)_+ \\ \sum_j w_{j|g} z_j = y}} \sum_{j \in G_g} w_{j|g} h_j(z_j).$$

The common terminal interpolant is $F_{g,c}$.

Proposition EC.5.1 (Exact class pooling). *For $c_1 \leq s \leq \sum_j w_{j|g} b_j$, the optimized class value is*

$$V_{g,c}(s) = \begin{cases} \sum_j w_{j|g} F_{g,c}(\theta_j(s)), & s \leq D_g(d), \\ \sum_j w_{j|g} F_{g,c}(\min(b_j, d)) - C_g(s - D_g(d); d), & s \geq D_g(d). \end{cases} \quad (7)$$

Both formulas agree at $D_g(d)$ and the value is continuous there. Relative to evaluating the canonical

response at $\theta(s)$, the exact correction is

$$\Delta_g(d; s) = \sum_{j \in G_g} \pi_j h_j((\theta_j(s) - d)_+) - W_g C_g((s - D_g(d))_+; d) \geq 0. \quad (8)$$

It depends on the book only through its last eligible selected level.

Proof. Below $D_g(d)$, Lemma 3.1 rules out intermediate service at an optimum. Capped equalization maximizes the sum of the same concave interpolant: among uncapped coordinates, equal levels satisfy the supporting-slope inequalities; capped smaller coordinates have no feasible upward movement, and the same common supporting slope supports their upper-bound constraint. This proves the first formula, also when the interpolant has affine pieces and the maximizing allocation is not unique.

Above $D_g(d)$, all terminal capacities must be filled and the residual is the stated capped-cost problem. At the junction, its resource and cost are zero, and $\theta_j(D_g(d)) = \min(b_j, d)$, so both expressions coincide. Continuity follows from continuity of interpolation and the finite capped convex value function. If $s > D_g(d)$, the equalization water level exceeds d ; otherwise its weighted mean would be at most $D_g(d)$. Thus $\theta_j(s) \geq \min(b_j, d)$ coordinatewise, and

$$\sum_j w_{j|g} (\theta_j(s) - d)_+ = s - D_g(d).$$

The equalized service vector is feasible for C_g , which proves (8). For $s \leq D_g(d)$ both service terms vanish. $\square$

For completeness, the fixed-target path to which this correction is applied is explicit. For a target vector t , put

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + (t_j - u)\sigma_{uv}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases}$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}.$$

Every target is owned by its unique bracketing selected edge, or by the terminal node. Hence its book value is the sum of these terms less the selected charges. A backward longest-path recurrence with a command counter optimizes the book in $O((k + m)N^2)$ operations.

Theorem EC.5.2 (Exact common-prefix joint design). *If $E = 1$, one fixed-target path calculation at $t_j = \theta_j(B)$, corrected by (8), solves (1) exactly. In the rational quadratic case its arithmetic cost is*

$$O((k + m)N^2 + kN \log(k + 1)).$$

It permits heterogeneous caps, heterogeneous costs including zero curvatures, rational nonnegative opening charges, and any feasible promise.

Proof. The sole class mean is fixed at B . For each possible last eligible level, Proposition EC.5.1 gives the exact difference between equalized-target evaluation and optimal class allocation. Along a selected path, add this correction once: to the edge that leaves the eligible catalog prefix, or to the terminal node if no selected edge leaves it. Every book is represented and corrected exactly once, without changing its charge or cardinality. Maximizing the corrected path values therefore maximizes the true joint objective. Compute the N candidate capped-cost corrections by rational water-level sweeps in $O(kN \log(k + 1))$ operations. The fixed-target path supplies the remaining bound. All input denominators and sorted rational events have polynomial encoding length. $\square$

For several classes the same construction is exact at *supplied* class means. The resource-path scheme avoids searching these means and therefore strengthens, rather than weakens, the variable-eligibility result. Exact enumeration remains polynomial for each fixed command budget but has an exponent depending on that budget. The new additive scheme is polynomial in the budget itself and inverse normalized accuracy. No claim of an exact unrestricted polynomial algorithm, a parameterized hardness lower bound, or a relative-error approximation follows from these statements.

EC.5.2 Insertion and threshold perturbation

Proposition EC.5.3 (Catalog and selected-book stability). *Insert one candidate into a catalog, leaving every original point and charge unchanged. The full-catalog class count changes from E to either E or $E + 1$; after r insertions it is at most $\min(k, E + r)$. For every retained selected book, its resource arcs and fixed-book value are unchanged. If no ceiling perturbation crosses a catalog candidate and each perturbed ceiling remains at least its cap, all feasible canonical responses and the global value are unchanged.*

Proof. Only the histories in the old catalog gap containing the inserted point can split an old class. They form at most one old class. Other classes retain their members and possibly acquire a new prefix-length integer. Thus there is at most one additional class per insertion. Fixed-book arc invariance is Corollary 3.5. If no ceiling crosses any candidate, each history has the same eligible prefix for every book; Proposition 2.1 then gives identical feasible targets and payoffs. $\square$

An insertion can therefore increase E without changing the optimum. Given an incumbent value v_0 , any new command with $\rho(z) > f(1) - v_0$ is dominated: a book using it has net value at most $f(1) - \rho(z) < v_0$, because all service costs and other charges are nonnegative. A dominated point inserted between two ceilings can split their full-catalog class, yet it cannot appear in an optimal book. The resource functions of every retained optimal path stay identical. A previously certified global upper bound must nevertheless be recomputed over the enlarged path set unless this dominance argument certifies exclusion of every added option.

Threshold margins matter. If $\min_{j,i} |\tau_j - a_i| > \delta$ and perturbations have size at most δ , with strict noncrossing at endpoints, the global value is invariant. Without a margin there need not be a vanishing error bound. For one history with $b = B = 1/4$, catalog $\{0, 1/2\}$, budget two, zero charges, $f(x) = 2x - x^2/2$, and $h(y) = y^2/2$, a ceiling just below $1/2$ gives value $-1/32$. At ceiling $1/2$, a half-and-half lottery gives $7/16$. The jump is $15/32$, although the ceiling change can be arbitrarily small.

For uncertain ceilings $\tau^- \leq \tau \leq \tau^+$, with $\tau^- \geq b$, monotonicity of the original feasible set gives

$$\underline{V}(\tau^-) \leq V^*(\tau) \leq \bar{V}(\tau^+). \quad (9)$$

The lower policy is feasible throughout the uncertainty set. Thus endpoint certificates give robust decisions without incorrectly assuming continuity across a command threshold. Post-optimization pooling may compress a fixed-book allocation certificate, but it must retain the original individual cap and service witnesses; it cannot certify omitted alternative books by itself.

EC.5.3 Safe command deletion and certificate transfer

An unchanged incumbent after catalog refinement does not prove that the enlarged problem has the same optimum. We give a sufficient exclusion test that controls every feasible allocation, rather than only the winning book. It is sharper than the gross bound $f(1) - \rho(z)$ and needs no globally optimal incumbent.

Assume the catalog is feasible, write $a = a_1$ and $\alpha = \min(B, \min_j b_j)$, and consider $z \in \mathcal{A}$ with $z > \alpha$. Such a command cannot be the anchor of a feasible book. Define

$$s_{az} = \frac{f(z) - f(a)}{z - a}, \quad (10)$$

$$G(z) = \sum_{j: \tau_j \geq z} \pi_j \{s_{az} \min(b_j - a, z - a) + h_j(b_j - a) - h_j((b_j - z)_+)\}. \quad (11)$$

All arguments are nonnegative because $a \leq \min_j b_j$. The expression depends on the candidate, the smallest catalog command, and the original histories, but not on other catalog entries.

Theorem EC.5.4 (Anchor-envelope screening). *For any feasible book containing a non-anchor command z , deleting z and reoptimizing its canonical responses at the same individual targets loses at most $G(z)$ in gross payoff. Consequently, every set*

$$S \subseteq \{z \in \mathcal{A} : z > \alpha, \rho(z) \geq G(z)\} \quad (12)$$

can be deleted simultaneously without changing $V^(B)$. At least one optimal book avoids S . If $\rho(z) > G(z)$, no optimal book contains z .*

The minimum catalog command is retained. Hence the bound for any remaining command is unchanged after screening or after insertion of other non-anchor commands that leave the minimum and the original primitives unchanged. In the rational quadratic specialization, computing all $G(z)$ takes $O(kN)$ arithmetic operations and polynomial bit complexity. This preprocessing does not change the command budget or the accuracy guarantee of Theorem EC.4.2.

Proof. First fix the immediate selected predecessor $u < z$. If the book has a selected successor, denote it by $v > z$; otherwise write $v = \dagger$. Histories with $\tau_j < z$ are unaffected by the deletion. For a remaining history, put $x_j = (b_j - u)_+$.

If $z \leq \tau_j < v$, or $v = \dagger$, deleting z changes the last eligible selected level from z to u . Below u the canonical response is unchanged. Between u and z , the gain from retaining z is $s_{uz}(t - u) + h_j(t - u)$, which

is nondecreasing. Above z the gain is $f(z) - f(u) + h_j(t - u) - h_j(t - z)$. Convexity makes the last cost increment nondecreasing in t . The largest gain over the target interval therefore occurs at b_j and equals

$$s_{uz} \min(x_j, z - u) + h_j(x_j) - h_j((b_j - z)_+). \quad (13)$$

This formula also gives zero when $b_j \leq u$.

If $\tau_j \geq v$ for a finite successor, the two canonical responses differ only on $[u, v]$. Their difference is the triangular interpolation gain, with height

$$d_f(u, z, v) = f(z) - \frac{v - z}{v - u} f(u) - \frac{z - u}{v - u} f(v) \geq 0$$

at z . Its maximum under the individual cap is

$$d_f(u, z, v) \min\{1, x_j/(z - u)\}. \quad (14)$$

Because f is increasing, $d_f(u, z, v) \leq f(z) - f(u)$. Thus (14) is no larger than (13); the additional cost difference in the latter is nonnegative. A terminal successor therefore bounds every successor context.

For fixed z , concavity of f makes s_{uz} nonincreasing as u increases. Both $\min((b_j - u)_+, z - u)$ and $h_j((b_j - u)_+)$ are also nonincreasing. The subtracted term $h_j((b_j - z)_+)$ is independent of u . Therefore (13) is largest at the smallest catalog predecessor a . Summing the resulting inequalities proves (11) as a uniform loss bound.

Deletion keeps the old anchor, and hence the old target vector remains feasible by Proposition 2.1. Canonical reconstruction satisfies every realization ceiling and expected cap, and preserves the weighted equality exactly. The opening charge saved is at least the gross loss whenever (12) holds. Apply this argument successively to each selected element of S . The minimum a is never deleted, so the same bounds remain valid after every deletion. Every original feasible value is weakly improved by a retained-catalog policy, whereas every retained-catalog policy was already feasible in the original catalog. The optima are equal. Strict inequality gives a strict improvement for any book containing that command, proving the final exclusion claim. Evaluating the rational formula once for each history and candidate gives the stated complexity. $\square$

The envelope is sharp as a bound over the individual target box: at $B = \bar{b}$, the books $\{a, z\}$ and $\{a\}$ attain exactly the gross difference $G(z)$. At smaller promises it can be conservative. Equality of a charge and its bound certifies existence of a retained optimum, not that every optimum avoids the command. For example, one history with $b = B = 1/4$, zero service cost, catalog $\{0, 1/2\}$, reward $2x - x^2/2$, and charge $\rho(1/2) = 7/16$ has both books optimal at value zero. Screening preserves the singleton optimum. The non-anchor condition is essential and is not replaced by an empirical claim that a command happened to be unused.

Corollary EC.5.5 (Refinement-safe global intervals). *Let a reduced catalog be obtained by the certified deletions in Theorem EC.5.4. An original-space feasible lower policy and a valid global upper bound for the reduced problem certify the same interval for the original problem. In particular, a previously checked catalog certificate remains valid after non-anchor insertions when every inserted option is excluded by the*

theorem, all retained charges and original primitives are unchanged, and the same promise and budget are used.

Proof. Theorem EC.5.4 gives equality of the two optimal values. The lower policy uses only retained commands and remains feasible in the enlarged catalog with the same payoff. The upper bound applies to the identical optimum. These comparisons preserve the interval width and therefore its requested accuracy. $\square$

The transfer certificate records the original instance, deleted candidates and their rational bounds, the reduced-catalog global certificate, and the original lower policy. The independent checker evaluates the possible predecessor–successor contexts at canonical breakpoints, rather than importing the screening formula. It checks their maximum against the declared bound, binds the reduced problem to unchanged original primitives, and invokes the separate resource-path checker. An externally supplied instance digest identifies the problem that the verifier intended to certify; a self-reported hash alone does not establish that identity. Neither a selected-book explanation nor a solver completion status can replace this global argument.

EC.6 Purpose and Reading Guide

This companion supplies the retained class-box support details, an explicit dyadic count, the independently implemented certificate arithmetic, and complete computational diagnostics. The principal selected-boundary reduction and capacity-repair proofs are in the article. No empirical calibration is asserted. The historical derivations are preserved in their original files rather than accumulated into this companion as additional prerequisites.

EC.7 The Retained Class-Box Price Identity

Fix a catalog-eligibility class, its last eligible selected level d , normalized weights w_j , and common mean interval $[\ell, u]$. Write $D = \sum_j w_j \min(b_j, d)$ and $c_j = (b_j - d)_+$. Let $\theta_j(s) = \min(b_j, z(s))$ be capped equalization. Set

$$a_j = (\theta_j(\ell) - d)_+, \quad b'_j = (\theta_j(u) - d)_+, \quad q_- = (\ell - D)_+, \quad q_+ = (u - D)_+.$$

When $u \geq D$, the endpoint identities needed for the service correction are

$$\sum_j w_j b'_j = u - D, \quad \sum_j w_j a_j = (\ell - D)_+. \quad (15)$$

Indeed, if $s \geq D$, its water level is at least d , and $\theta_j(s) = \min(b_j, d) + (\theta_j(s) - d)_+$. Summation proves the identity. If $s < D$, the water level is at most d and all positive parts vanish. Thus every independently bounded service vector in $\prod_j [a_j, b'_j]$ has aggregate service in $[q_-, q_+]$. This feasibility inclusion, not merely concavity, is what proves the correction's sign.

For a price λ , define

$$T(\lambda) = \max_{\substack{0 \leq y_j \leq c_j \\ q_- \leq \sum_j w_j y_j \leq q_+}} \sum_j w_j [-h_j(y_j) - \lambda y_j], \quad (16)$$

$$J(\lambda) = \sum_j w_j \max_{a_j \leq y \leq b'_j} [-h_j(y) - \lambda y]. \quad (17)$$

When $u \geq D$, the true class support minus the support over the artificial individual box $[\theta(\ell), \theta(u)]$ is $T - J$. The difference is nonnegative by (15). If $u < D$, both supports are terminal-only and the difference is zero.

Here is the full price-case argument. If $\lambda \leq 0$ and $u \geq D$, positive terminal reward minus price fills all available terminal capacity before service is selected. The true and artificial problems then share terminal reward and its price constant, and their only difference is (16) minus (17). If $\lambda > 0$ and $\ell \leq D$, avoidable service strictly lowers the priced objective. Both problems have a terminal-only maximizer. At every attainable terminal mean, capped equalization maximizes the common concave interpolant and lies inside the artificial box; hence their suprema coincide. In this case $q_- = 0$, each $a_j = 0$, and both T and J are zero. If $\lambda > 0$ and $\ell > D$, service is unavoidable and both problems choose their smallest permitted mean. The terminal constants agree, and their service difference is again $T - J$. All equalities include $\lambda = 0$, $\ell = D$, $u = D$, zero costs, and zero residual capacities.

For the quadratic service cost, a multiplier ν gives the upper witness

$$T(\lambda) \leq \nu \begin{cases} q_+, & \nu \geq 0, \\ q_-, & \nu < 0 \end{cases} + \sum_j w_j \max_{0 \leq y \leq c_j} \{-\gamma_j y^2 / 2 - (\lambda + \nu)y\}. \quad (18)$$

For positive curvature the scalar maximizer is the stationary point clipped to $[0, c_j]$; for zero curvature one chooses an endpoint or the entire interval when its slope is zero. Equality of this upper witness with a feasible vector verifies T without trusting the allocation routine. The inherited checker also verifies capped equalization, the sign of the correction, and its ownership by exactly one crossing edge or terminal node. Its original two-sided path tables and complete class-box cover remain in the source archive and are rerun for the matched class-subdivision study. They are not used to justify the new resource-path upper bound.

EC.8 An Explicit Dyadic Bound for the Retained Class-Mean Method

The resource-path guarantee does not require class-mean subdivision. We nevertheless give an explicit bound for that retained method, both for comparison and to remove ambiguity about its stopping scale.

Let $X = [\ell, u]$ be a projected class-mean box intersecting $\sum_g W_g s_g = B$. Fix a dependent class g_0 and let $d = E - 1 > 0$. A zero-price box oracle supplies an upper value. Its selected book is feasible at the original promise; bounded one-direction transfers of its class means restore equality. Since every fixed-book class response is L -Lipschitz,

$$\text{zero-price upper value} - \text{recovered lower value} \leq L \sum_g W_g (u_g - \ell_g).$$

Projection onto the root equality gives

$$W_{g_0}(u_{g_0} - \ell_{g_0}) \leq \sum_{g \neq g_0} W_g(u_g - \ell_g).$$

Thus the explicit free-coordinate stopping threshold

$$\eta_{\text{box}} = \frac{\varepsilon}{2L(E-1)} \quad (19)$$

suffices: if every free weighted width is at most η_{box} , the gap is at most ε . The zero price must be included even when additional prices are used.

Proposition EC.8.1 (Dyadic cover, with the ceiling retained). *Let $w_g^0 = W_g(u_g^0 - \ell_g^0)$ be the initial free weighted widths. Bisect a largest free weighted width at every unresolved node, projecting children onto the equality. Define*

$$q_g = \begin{cases} 0, & w_g^0 = 0, \\ \max\{0, \lceil \log_2(w_g^0/\eta_{\text{box}}) \rceil\}, & w_g^0 > 0. \end{cases}$$

A complete binary cover has at most

$$M \leq \prod_{g \neq g_0} 2^{q_g} \leq \left(\max \left\{ 1, \frac{4L(E-1)}{\varepsilon} \right\} \right)^{E-1} \quad (20)$$

leaves, and at most $2M - 1$ nodes when both children of every split are counted. The first product is a dyadic upper bound, not a claim that projection must generate every tensor-grid leaf.

Proof. A node with gap above ε has some free width greater than η_{box} . Its largest free width also exceeds this threshold. A split halves that coordinate's width, while projection never increases any width. Coordinate g can therefore be split at most q_g times on a root-to-leaf path. Every path has depth at most $\sum_g q_g$, so its binary tree has at most $2^{\sum_g q_g}$ leaves. A full binary tree with M leaves has $2M - 1$ nodes. If infeasible children are pruned, including them before pruning provides the same upper bound.

For the coarse bound, each $w_g^0 \leq 1$. If $\eta_{\text{box}} \geq 1$, all q_g vanish. Otherwise, $2^{\lceil \log_2(1/\eta_{\text{box}}) \rceil} \leq 2/\eta_{\text{box}} = 4L(E-1)/\varepsilon$. Multiplication proves (20). $\square$

For width one and threshold $1/3$, the dyadic requirement is four leaves, not three. The coarse bound must be evaluated with the *same* threshold: (19) then gives $4L(E-1)/\varepsilon = 6$ in one free dimension. Conversely, setting $4L/\varepsilon = 3$ gives $\eta_{\text{box}} = 2/3$ and requires only two leaves. These examples distinguish an omitted ceiling in a putative exact product from a valid coarser bound that already includes its factor of two. They also explain why the stopping threshold must not silently change from $\varepsilon/(2Ld)$ to $\varepsilon/(4Ld)$.

A repaired class-mean net remains valid as well. For each feasible anchor, round $E - 1$ weighted class means down with step $\varepsilon/(2L(E-1))$, set the final coordinate by the equality, clip and repair the remaining deficit by bounded transfers. The weighted absolute perturbation is at most twice the rounded mass. Exact conditional class optimization then loses at most ε . This yields at most $N(1 + 2L(E-1)/\varepsilon)^{E-1}$ calls.

Its class-dependent exponent is a property of this particular enumeration, not a lower bound on the joint problem; Theorem EC.4.2 removes that exponent.

EC.9 Independent Resource-Path Verification

EC.9.1 Marginal intervals and exact common denominators

For one service exit set, all zero-curvature capacity can be used at zero cost. Let that weighted mass be q_0 . For every remaining history, the marginal saturation threshold is $\gamma_j c_j$. Sort distinct thresholds, including zero. On an interval between consecutive thresholds, let S be the saturated set and A the active set. The mass, reciprocal curvature, and saturated cost are

$$y_0 = q_0 + \sum_{j \in S} \pi_j c_j, \quad R_0 = \sum_{j \in A} \frac{\pi_j}{\gamma_j}, \quad c_0 = \sum_{j \in S} \frac{\pi_j \gamma_j c_j^2}{2}.$$

At required mass y , set $\alpha = (y - y_0)/R_0$. If α lies in the interval, the allocation has $z_j = c_j$ on S , $z_j = \alpha/\gamma_j$ on A , and the cost is $c_0 + R_0 \alpha^2/2$. These first-order conditions support the convex objective globally. At an endpoint, adjacent formulas agree. At total capacity the final formula is valid; a set containing only free capacity has cost zero throughout and needs no active interval.

The new optimizer evaluates capped allocation through the antecedent water-filling routine. The new checker instead builds these marginal intervals directly from the model and evaluates the required mass independently. It recomputes every terminal capacity and every exit set. It checks that the supplied common denominator divides every rational sample exactly and that the integer sample sequence has nonincreasing differences. No floating-point operation is used in the acceptance decision.

The denominator is an exact scaling device, not an accuracy parameter. Its bit length can be large because it is the least common multiple across many distinct rational samples. The polynomial bit bound in the article counts that length explicitly. Reporting only the number of grid states would hide this arithmetic cost.

EC.9.2 Why finite padding preserves the row maxima

Let a previous Bellman vector have finite entries $A_0, \dots, A_R$, and a concave kernel have entries $K_0, \dots, K_C$. Only rows $0 \leq z \leq R + C$ are relevant. Each such row has a feasible column with $0 \leq z - t \leq C$. Set

$$M = \max_t |A_t| + \max_q |K_q|, \quad P = 4M + 1.$$

Extend the kernel to negative indices by $K_0 + Pq$, and beyond C by $K_C - P(q - C)$. These extensions have slopes larger than every internal forward difference on the left and smaller on the right, so the extended sequence remains concave. An infeasible entry is at most $M - P < -M$, whereas a feasible entry is at least $-M$. Thus no infeasible entry is a row maximizer. The extended matrix $A_t + K_{z-t}$ is finite and anti-Monge; its leftmost row maximizers are totally monotone. The checker applies the standard matrix-search reduction of Aggarwal et al. (1987) to this rectangular matrix and verifies that every returned index lies inside the

original feasible band.

The optimizer uses a different divide-and-conquer monotone search. Agreement between the two implementations is tested against direct convolution on two thousand seeded arrays with arbitrary previous values and concave kernels. Concavity of the previous Bellman vector is never assumed.

EC.9.3 Coverage and original-space feasibility

For each feasible anchor, the certificate must supply exactly the expected command-count and terminal-vertex state sets. The checker reconstructs all transitions, compares each complete row, computes all terminal completions, and checks the accepted resource band. Missing anchors or shortened rows fail even if the final displayed bound happens to look plausible. The best score across this complete set determines the upper bound; a purported solver status does not.

The lower witness retains every original history. For each history, the checker verifies nonnegative intermediate service, support inside the selected book, nonnegative probabilities summing to one, every positive-probability realization ceiling, the expected cap, and the reported target. It then checks the exact weighted root promise and subtracts all selected charges once. It validates the interval and requested tolerance using rational arithmetic. Pooling or compressing a display does not remove these individual witnesses.

The corruption suite changes, one at a time, the schema, an anchor, a Bellman row, the upper bound, the lower bound, the best grid score, the common denominator, the promise, the Lipschitz constant, the requested tolerance, service nonnegativity, and a lottery mass. Every mutation is rejected. Assertions implement verification failures; Python must not be invoked with optimization flags that disable assertions. The reproduction scripts enforce ordinary Python execution.

EC.10 A Second Model: Compatible Production Modules

The capacity-repair theorem is not specific to renewal lotteries. Consider a production interface assembled from compatible modules. A directed acyclic graph describes compatibility; choosing a source-to-sink path installs its modules, with at most h modules and installation payment $\kappa_e \geq 0$ for module e . Module e can process a divisible amount $x_e \in [0, C_e]$ and earns concave net operating return $\psi_e(x_e)$. An accepted order of size R must be assigned exactly:

$$\max_{P,x} \left\{ \sum_{e \in P} [\psi_e(x_e) - \kappa_e] : |P| \leq h, \sum_{e \in P} x_e = R, 0 \leq x_e \leq C_e \right\}. \quad (21)$$

Modules on a selected compatibility chain supply parallel divisible capacity; the order is allocated across them, not passed serially through each module. This distinguishes the workload equality in (21) from a serial-flow bottleneck model.

One relevant design is capacity-conservative configuration. Each graph vertex has a cumulative capacity coordinate p_v , and a module spanning u to v supplies $C_{uv} = p_v - p_u \geq 0$. A complete chain covers the same total capacity regardless of how finely it is configured. The operating curves and installation charges may

differ across compatible modules. For example, $\psi_e(x) = \alpha_e x - \beta_e x^2/2$, with $\beta_e \geq 0$, represents diminishing operating return; it need not be increasing over the entire capacity range.

Corollary EC.10.1 (Original-order production guarantee). *Suppose every admissible configuration in (21) has total capacity at least R , and all operating curves are L -Lipschitz. Resource gridding with step $\eta = \varepsilon/(2Lh)$ and residual repair returns a feasible configuration and an additive interval of width at most ε , without exceeding the module budget. If the curves are concave and rational grid values can be evaluated in polynomial time, the length-indexed dynamic program has $O(h|\mathcal{E}|Q \log(Q+1))$ arithmetic complexity after preprocessing, where $Q = \lceil R/\eta \rceil + 1$.*

Proof. Every accepted rounded configuration can absorb its omitted workload by the capacity assumption. Theorem EC.4.1 therefore applies directly. A topological dynamic program uses the same scalar-resource convolution as (3), now over graph arcs rather than catalog pairs and without anchor enumeration. Concavity gives the monotone convolution bound. Capacity-conservative configurations satisfy the assumption by telescoping $\sum_{e \in P} C_e = p_{\text{sink}} - p_{\text{source}}$. $\square$

This second model has neither heterogeneous renewal histories nor terminal-command lotteries. It reuses the capacity-repair mechanism, whereas Theorem 3.4 derives that mechanism from the original renewal constraints. We offer the model as a mathematical application of the abstraction, not as an observed industrial deployment. Empirical calibration of either setting requires operational data beyond the synthetic study reported here.

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Table EC.1: Complete numerical mixed-integer diagnostics on the historical cases.

Seed	Envelope	Variables	Binary	Nodes	Solver s	Limit s	Error	Width
49500	tangent	249	117	1	0.0219	2	1.1e-04	0
49500	secant	249	117	1	0.0171	2	1.1e-04	0
49501	tangent	249	117	1	0.0174	2	1.3e-04	0
49501	secant	249	117	1	0.0154	2	1.3e-04	0
49502	tangent	249	117	1	0.0075	2	4.3e-05	5.6e-17
49502	secant	249	117	1	0.0080	2	4.3e-05	5.6e-17
49503	tangent	249	117	1	0.0228	2	5.9e-05	-1.4e-17
49503	secant	249	117	1	0.0132	2	5.9e-05	-1.4e-17
49504	tangent	249	117	1	0.0085	2	2.8e-05	0
49504	secant	249	117	1	0.0070	2	2.8e-05	0
49505	tangent	249	117	1	0.0109	2	8.8e-05	0
49505	secant	249	117	1	0.0099	2	8.8e-05	0

Solver seconds are actual per-envelope calls; each solver limit is two seconds, within the total four-second algorithm alarm. Error is the analytical service-envelope approximation allowance, not a time allowance. Width is the signed floating-point difference of the two returned numerical values; tiny negative values are roundoff and are not recast as exact certificates. All raw primal/dual values, status codes, relative solver gaps and residuals are retained in the machine-readable records.

Table EC.2: Target-command threshold slacks and exact relevance in every screening case.

Family	Ratio	z	$G(z)$	Charge	Slack	Envelope	Forced
0	9/10	0.50	0.7384	0.6646	-0.0738	0	1
0	1	0.50	0.7384	0.7384	0	1	1
0	11/10	0.50	0.7384	0.8122	0.0738	1	1
1	9/10	0.75	0.4652	0.4187	-0.0465	0	1
1	1	0.75	0.4652	0.4652	0	1	1
1	11/10	0.75	0.4652	0.5117	0.0465	1	1
2	9/10	0.50	0.7140	0.6426	-0.0714	0	1
2	1	0.50	0.7140	0.7140	0	1	1
2	11/10	0.50	0.7140	0.7854	0.0714	1	1
3	9/10	0.75	0.5916	0.5324	-0.0592	0	1
3	1	0.75	0.5916	0.5916	0	1	1
3	11/10	0.75	0.5916	0.6508	0.0592	1	1
4	9/10	0.50	0.8006	0.7205	-0.0801	0	1
4	1	0.50	0.8006	0.8006	0	1	1
4	11/10	0.50	0.8006	0.8806	0.0801	1	1
5	9/10	0.75	0.5421	0.4879	-0.0542	0	1
5	1	0.75	0.5421	0.5421	0	1	1
5	11/10	0.75	0.5421	0.5963	0.0542	1	1
6	9/10	0.50	0.7463	0.6717	-0.0746	0	1
6	1	0.50	0.7463	0.7463	0	1	1
6	11/10	0.50	0.7463	0.8210	0.0746	1	1
7	9/10	0.75	0.8500	0.7650	-0.0850	0	1
7	1	0.75	0.8500	0.8500	0	1	1
7	11/10	0.75	0.8500	0.9351	0.0850	1	1

Each family uses the same nonuniform caps, promise and mixed catalog at charge ratios 0.9, 1, 1.1. The full command records additionally give G , charge, slack, forced upper bound and exact inclusion/exclusion value for every candidate, not just this target. A below-envelope charge does not by itself prove usefulness. The separate sharpness examples below demonstrate that usefulness can change across the equality threshold.

Table EC.3: Exact sharpness neighbors and a distinct divisible-production example.

Charge ratio	Envelope	Charge	Singleton	Pair	Pair needed
9/10	0.785000	0.706500	-0.146667	-0.068167	1
1	0.785000	0.785000	-0.146667	-0.146667	0
11/10	0.785000	0.863500	-0.146667	-0.225167	0

Module budget	Order	Exact value	Supporting price	Selected vertices
1	3	27/4	3/2	0,3
1	6	9	0	0,3
1	8	8	-1	0,3
2	3	181/16	13/4	0,1,3
2	6	39/2	9/4	0,2,3
2	8	93/4	3/2	0,2,3
3	3	685/52	57/13	0,1,2,3
3	6	97/4	3	0,1,2,3
3	8	799/28	9/7	0,1,2,3

The sharpness configuration has $B = \bar{b}$, nonuniform caps and a feasible two-command budget; equality allows both books. Production uses vertices of capacities 0, 2, 5, 9 and six independently specified compatible modules, with parallel divisible order assignment. Exact allocations and every alternative configuration are recorded. This is a numerical mathematical example, not an industrial calibration or a renamed history/lottery instance.